

PARCIAL 2 RdC

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Carnet: 00097524

A: 00

B: 24

EJERCICIO 1-Estatico-Red base asignada: 10.24.0.0/24

Subneteo:

Walter Samuel Guillen Gonzalez - 00097524 B=24 A=00

Ejercicio 1: VLSM - Estático Red = 10.24.0.0/24

Red	Subneteo	Ventilador	Superficie	Gerencia	Enlaces
Admin 4	Admin	Ventilador	Superficie	Gerencia	Enlaces
Ventilador 80	2 ⁸ - 277/10	2 ⁸ - 277/10	2 ⁸ - 277/10	2 ⁸ - 277/10	2 ⁸ - 277/10
Superficie 14	2 ⁵ - 2 = 62	2 ⁵ - 2 = 30	2 ⁵ - 2 = 14	2 ⁵ - 2 = 14	2 ⁵ - 2 = 2
Gerencia 10	32 - 6 = 26	32 - 5 = 27	32 - 4 = 28	32 - 4 = 28	32 - 2 = 30
R1-R2 2	255.255.255.192	255.255.255.224	255.255.255.240	255.255.255.246	255.255.255.252
R2-R3 2	256 - 192 = 64	256 - 224 = 32	256 - 240 = 16	256 - 246 = 10	256 - 252 = 4

(3) Tabla VLSM

Red	Rango	Broadcast
Admin	10.24.0.1 - 10.24.0.63	10.24.0.63
Ventilador	10.24.0.64 - 10.24.0.95	10.24.0.95
Superficie	10.24.0.96 - 10.24.0.111	10.24.0.111
Gerencia	10.24.0.112 - 10.24.0.127	10.24.0.127
R1-R2	10.24.0.128 - 10.24.0.131	10.24.0.131
R2-R3	10.24.0.132 - 10.24.0.135	10.24.0.135

(4) Tabla GNS3

Red	Rango	Broadcast
Admin	10.24.0.1 - 10.24.0.63	10.24.0.63
Ventilador	10.24.0.64 - 10.24.0.95	10.24.0.95
Superficie	10.24.0.96 - 10.24.0.111	10.24.0.111
Gerencia	10.24.0.112 - 10.24.0.127	10.24.0.127
R1-R2	10.24.0.128 - 10.24.0.131	10.24.0.131
R2-R3	10.24.0.132 - 10.24.0.135	10.24.0.135

Configuracion ip de cada pc:

```
PC1
PC2
PC3
PC4

For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC1> ip 10.24.0.2 255.255.255.192 10.24.0.1
Checking for duplicate address...
savePC1 : 10.24.0.2 255.255.255.192 gateway 10.24.0.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.24.0.2/26
GATEWAY    : 10.24.0.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC1> █
```

```
PC1
PC2
PC3
PC4

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC2> ip 10.24.0.66 255.255.255.224 10.24.0.65
Checking for duplicate address...
PC1 : 10.24.0.66 255.255.255.224 gateway 10.24.0.65

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.24.0.66/27
GATEWAY    : 10.24.0.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC2> █
```

```
PC1 PC2 PC3 PC4
Executing the startup file

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC3> ip 10.24.0.98 255.255.255.240 ip 10.24.0.97
Invalid options

PC3> ip 10.24.0.98 255.255.255.240 10.24.0.97
Checking for duplicate address...
PC1 : 10.24.0.98 255.255.255.240 gateway 10.24.0.97

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 10.24.0.98/28
GATEWAY    : 10.24.0.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC3> █
```

```
PC1 PC2 PC3 PC4
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500

PC4> ip 10.24.0.114 255.255.255.240 10.24.0.113
Checking for duplicate address...
PC1 : 10.24.0.114 255.255.255.240 gateway 10.24.0.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 10.24.0.114/28
GATEWAY    : 10.24.0.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500

PC4> █
```

Configuracion de interfaces en los routers:

```
PC1 PC2 PC3 PC4 R1 R2 R3
FastEthernet0/1 unassigned YES unset administratively down down
FastEthernet1/0 unassigned YES unset administratively down down
R1-00097524#enable
R1-00097524#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1-00097524(config)#interface fastEthernet0/0
R1-00097524(config-if)#ip address 10.24.0.1 255.255.255.192
R1-00097524(config-if)#no shutdown
R1-00097524(config-if)#exit
R1-00097524(config)#
*Mar 1 00:05:07.083: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:05:08.083: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1-00097524(config)#interface fastEthernet0/1
R1-00097524(config-if)#ip address 10.24.0.65 255.255.255.224
R1-00097524(config-if)#no shutdown
R1-00097524(config-if)#exit
R1-00097524(config)#interfa
R1-00097524(config)#
*Mar 1 00:05:38.531: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:05:39.531: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1-00097524(config)#interface fastEthernet1/0
R1-00097524(config-if)#ip address 10.24.0.129 255.255.255.252
R1-00097524(config-if)#no shutdown
R1-00097524(config-if)#exit
R1-00097524(config)#
*Mar 1 00:06:12.911: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:06:13.911: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1-00097524(config)#end
R1-00097524#write memory
*Mar 1 00:06:16.915: %SYS-5-CONFIG_I: Configured from console by console
R1-00097524#write memory
Building configuration...
[OK]
R1-00097524#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.24.0.1 YES manual up
FastEthernet0/1 10.24.0.65 YES manual up
FastEthernet1/0 10.24.0.129 YES manual up
R1-00097524#
```

```
PC1 PC2 PC3 PC4 R1 R2 R3
FastEthernet0/1 unassigned YES unset administratively down down
FastEthernet1/0 unassigned YES unset administratively down down
R2-00097524#enable
R2-00097524#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2-00097524(config)#interface fastEthernet0/0
R2-00097524(config-if)#ip address 10.24.0.97 255.255.255.240
R2-00097524(config-if)#no shutdown
R2-00097524(config-if)#exit
R2-00097524(config)#
*Mar 1 00:06:50.751: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:51.751: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2-00097524(config)#interface fastEthernet0/1
R2-00097524(config-if)#ip address 10.24.0.130 255.255.255.252
R2-00097524(config-if)#no shutdown
R2-00097524(config-if)#exit
R2-00097524(config)#interface fa
R2-00097524(config)#
*Mar 1 00:07:21.599: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:07:22.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2-00097524(config)#interface fastEthernet1/0
R2-00097524(config-if)#ip address 10.24.0.133 255.255.255.252
R2-00097524(config-if)#no shutdown
R2-00097524(config-if)#exit
R2-00097524(config)#en
R2-00097524(config)#
*Mar 1 00:07:50.987: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:07:51.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2-00097524(config)#end
R2-00097524#writ
R2-00097524#
*Mar 1 00:07:53.587: %SYS-5-CONFIG_I: Configured from console by console
R2-00097524#write memory
Building configuration...
[OK]
R2-00097524#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.24.0.97 YES manual up
FastEthernet0/1 10.24.0.130 YES manual up
FastEthernet1/0 10.24.0.133 YES manual up
R2-00097524#
```

```
PC1 PC2 PC3 PC4 R1 R2 R3
R3-00097524(config)#end
R3-00097524#write
*Mar 1 00:04:04.187: %SYS-5-CONFIG_I: Configured from console by console
R3-00097524#write memory
Building configuration...
[OK]
R3-00097524#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 unassigned YES unset administratively down down
FastEthernet0/1 unassigned YES unset administratively down down
R3-00097524#enable
R3-00097524#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3-00097524(config)#interface fastethernet0/0
R3-00097524(config-if)#ip address 10.24.0.113 255.255.255.240
R3-00097524(config-if)#no shutdown
R3-00097524(config-if)#exit
R3-00097524(config)#interface faste
*Mar 1 00:09:35.131: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:09:36.131: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3-00097524(config)#interface fastethernet0/1
R3-00097524(config-if)#ip address 10.24.0.134 255.255.255.252
R3-00097524(config-if)#no shutdown
R3-00097524(config-if)#exit
R3-00097524(config)#end
*Mar 1 00:09:59.579: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:10:00.579: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3-00097524(config)#end
R3-00097524#write
*Mar 1 00:10:01.615: %SYS-5-CONFIG_I: Configured from console by console
R3-00097524#write memory
Building configuration...
[OK]
R3-00097524#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.24.0.113 YES manual up up
FastEthernet0/1 10.24.0.134 YES manual up up
R3-00097524#
```

Configuracion de rutas estáticas :

```
PC1 PC2 PC3 PC4 R1 R2 R3
Building configuration...
[OK]
R1-00097524#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.24.0.1 YES manual up up
FastEthernet0/1 10.24.0.65 YES manual up up
FastEthernet1/0 10.24.0.129 YES manual up up
R1-00097524#enable
R1-00097524#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1-00097524(config)#ip route 10.24.0.96 255.255.255.240 10.24.0.130
R1-00097524(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.130
R1-00097524(config)#ip route 10.24.0.132 255.255.255.252 10.24.0.130
R1-00097524(config)#end
R1-00097524#writ
*Mar 1 00:14:20.139: %SYS-5-CONFIG_I: Configured from console by console
R1-00097524#write memory
Building configuration...
[OK]
R1-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C    10.24.0.0/26 is directly connected, FastEthernet0/0
C    10.24.0.64/27 is directly connected, FastEthernet0/1
S    10.24.0.112/28 [1/0] via 10.24.0.130
S    10.24.0.96/28 [1/0] via 10.24.0.130
C    10.24.0.128/30 is directly connected, FastEthernet1/0
S    10.24.0.132/30 [1/0] via 10.24.0.130
R1-00097524#
```

```
Building configuration...
[OK]
R2-00097524#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0     10.24.0.97      YES manual up          up
FastEthernet0/1     10.24.0.130     YES manual up          up
FastEthernet1/0     10.24.0.133     YES manual up          up
R2-00097524#enable
R2-00097524#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2-00097524(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.129
R2-00097524(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.129
R2-00097524(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.134
R2-00097524(config)#end
R2-00097524#write mem
*Mar 1 00:17:18.671: %SYS-5-CONFIG_I: Configured from console by console
R2-00097524#write memory
Building configuration...
[OK]
R2-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.129
S       10.24.0.64/27 [1/0] via 10.24.0.129
S       10.24.0.112/28 [1/0] via 10.24.0.134
C       10.24.0.96/28 is directly connected, FastEthernet0/0
C       10.24.0.128/30 is directly connected, FastEthernet0/1
C       10.24.0.132/30 is directly connected, FastEthernet1/0
R2-00097524#
```

```
Building configuration...
[OK]
R3-00097524#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0     10.24.0.113     YES manual up          up
FastEthernet0/1     10.24.0.134     YES manual up          up
R3-00097524#enable
R3-00097524#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3-00097524(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.133
R3-00097524(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.133
R3-00097524(config)#ip route 10.24.0.96 255.255.255.240 10.24.0.133
R3-00097524(config)#ip route 10.24.0.128 255.255.255.252 10.24.0.133
R3-00097524(config)#end
R3-00097524#wr
*Mar 1 00:20:04.907: %SYS-5-CONFIG_I: Configured from console by console
R3-00097524#write memory
Building configuration...
[OK]
R3-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.133
S       10.24.0.64/27 [1/0] via 10.24.0.133
C       10.24.0.112/28 is directly connected, FastEthernet0/0
S       10.24.0.96/28 [1/0] via 10.24.0.133
S       10.24.0.128/30 [1/0] via 10.24.0.133
C       10.24.0.132/30 is directly connected, FastEthernet0/1
R3-00097524#
```

Prueba de conexión exitosa:

```
PC1> show ip
NAME       : PC1[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC1> ip 10.24.0.2 255.255.255.192 10.24.0.1
Checking for duplicate address...
savePC1 : 10.24.0.2 255.255.255.192 gateway 10.24.0.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip
NAME       : PC1[1]
IP/MASK    : 10.24.0.2/26
GATEWAY    : 10.24.0.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC1> trace 10.24.0.114
trace to 10.24.0.114, 8 hops max, press Ctrl+C to stop
 1  10.24.0.1   15.022 ms  15.264 ms  15.148 ms
 2   *10.24.0.130 46.581 ms  45.101 ms
 3   *10.24.0.134 75.076 ms  76.113 ms
 4   **10.24.0.114 93.299 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>
```

EJERCICIO 2-Dinamico-Red base asignada: 192.24.0.0/24

Subneteo:

Wells Samuel Guillon Gonzalez - 00097524 B=24 A=00

Ejercicio 2 - VSIIM - Dinamico Red = 192.24.0.0/24

① Orde

Lon PC1 = 40 ✓

Lon PC2 = 28 ✓

Lon PC3 = 18 ✓

Lon PC4 = 9 ✓

Lon PC5 = 5 ✓

R1-R2 = 2 ✓

R1-R3 = 2 ✓

R2-R3 = 2 ✓

R3-R4 = 2 ✓

R4-R5 = 2 ✓

③ Table Vsm

Red	Primer host	Ultimo host	Broadcast
Lon PC1			
192.24.0.0/26	192.24.0.1	192.24.0.62	192.24.0.63
Lon PC2			
192.24.0.64/27	192.24.0.65	192.24.0.94	192.24.0.95
Lon PC3			
192.24.0.96/27	192.24.0.97	192.24.0.126	192.24.0.127
Lon PC4			
192.24.0.128/28	192.24.0.129	192.24.0.142	192.24.0.143
Lon PC5			
192.24.0.144/29	192.24.0.145	192.24.0.150	192.24.0.151
R1-R2			
192.24.0.152/30	192.24.0.153	192.24.0.154	192.24.0.155
R2-R3			
192.24.0.156/30	192.24.0.157	192.24.0.158	192.24.0.159
R2-R5			
192.24.0.160/30	192.24.0.161	192.24.0.162	192.24.0.163
R3-R4			
192.24.0.164/30	192.24.0.165	192.24.0.166	192.24.0.167
R4-R5			
192.24.0.168/30	192.24.0.169	192.24.0.170	192.24.0.171

③ Subneteo

Lon PC1

$2^9 - 2 = 510$

$2^8 - 2 = 254$

$2^7 - 2 = 126$

$2^6 - 2 = 62$

$2^5 - 2 = 30$

$2^4 - 2 = 14$

$2^3 - 2 = 6$

$2^2 - 2 = 2$

$2^1 - 2 = 0$

$2^0 - 2 = -2$

$2^{-1} - 2 = -4$

$2^{-2} - 2 = -6$

$2^{-3} - 2 = -8$

$2^{-4} - 2 = -10$

$2^{-5} - 2 = -12$

$2^{-6} - 2 = -14$

$2^{-7} - 2 = -16$

$2^{-8} - 2 = -18$

$2^{-9} - 2 = -20$

$2^{-10} - 2 = -22$

$2^{-11} - 2 = -24$

$2^{-12} - 2 = -26$

$2^{-13} - 2 = -28$

$2^{-14} - 2 = -30$

$2^{-15} - 2 = -32$

$2^{-16} - 2 = -34$

$2^{-17} - 2 = -36$

$2^{-18} - 2 = -38$

$2^{-19} - 2 = -40$

$2^{-20} - 2 = -42$

$2^{-21} - 2 = -44$

$2^{-22} - 2 = -46$

$2^{-23} - 2 = -48$

$2^{-24} - 2 = -50$

$2^{-25} - 2 = -52$

$2^{-26} - 2 = -54$

$2^{-27} - 2 = -56$

$2^{-28} - 2 = -58$

$2^{-29} - 2 = -60$

$2^{-30} - 2 = -62$

$2^{-31} - 2 = -64$

$2^{-32} - 2 = -66$

$2^{-33} - 2 = -68$

$2^{-34} - 2 = -70$

$2^{-35} - 2 = -72$

$2^{-36} - 2 = -74$

$2^{-37} - 2 = -76$

$2^{-38} - 2 = -78$

$2^{-39} - 2 = -80$

$2^{-40} - 2 = -82$

$2^{-41} - 2 = -84$

$2^{-42} - 2 = -86$

$2^{-43} - 2 = -88$

$2^{-44} - 2 = -90$

$2^{-45} - 2 = -92$

$2^{-46} - 2 = -94$

$2^{-47} - 2 = -96$

$2^{-48} - 2 = -98$

$2^{-49} - 2 = -100$

$2^{-50} - 2 = -102$

$2^{-51} - 2 = -104$

$2^{-52} - 2 = -106$

$2^{-53} - 2 = -108$

$2^{-54} - 2 = -110$

$2^{-55} - 2 = -112$

$2^{-56} - 2 = -114$

$2^{-57} - 2 = -116$

$2^{-58} - 2 = -118$

$2^{-59} - 2 = -120$

$2^{-60} - 2 = -122$

$2^{-61} - 2 = -124$

$2^{-62} - 2 = -126$

$2^{-63} - 2 = -128$

$2^{-64} - 2 = -130$

$2^{-65} - 2 = -132$

$2^{-66} - 2 = -134$

$2^{-67} - 2 = -136$

$2^{-68} - 2 = -138$

$2^{-69} - 2 = -140$

$2^{-70} - 2 = -142$

$2^{-71} - 2 = -144$

$2^{-72} - 2 = -146$

$2^{-73} - 2 = -148$

$2^{-74} - 2 = -150$

$2^{-75} - 2 = -152$

$2^{-76} - 2 = -154$

$2^{-77} - 2 = -156$

$2^{-78} - 2 = -158$

$2^{-79} - 2 = -160$

$2^{-80} - 2 = -162$

$2^{-81} - 2 = -164$

$2^{-82} - 2 = -166$

$2^{-83} - 2 = -168$

$2^{-84} - 2 = -170$

$2^{-85} - 2 = -172$

$2^{-86} - 2 = -174$

$2^{-87} - 2 = -176$

$2^{-88} - 2 = -178$

$2^{-89} - 2 = -180$

$2^{-90} - 2 = -182$

$2^{-91} - 2 = -184$

$2^{-92} - 2 = -186$

$2^{-93} - 2 = -188$

$2^{-94} - 2 = -190$

$2^{-95} - 2 = -192$

$2^{-96} - 2 = -194$

$2^{-97} - 2 = -196$

$2^{-98} - 2 = -198$

$2^{-99} - 2 = -200$

$2^{-100} - 2 = -202$

$2^{-101} - 2 = -204$

$2^{-102} - 2 = -206$

$2^{-103} - 2 = -208$

$2^{-104} - 2 = -210$

$2^{-105} - 2 = -212$

$2^{-106} - 2 = -214$

$2^{-107} - 2 = -216$

$2^{-108} - 2 = -218$

$2^{-109} - 2 = -220$

$2^{-110} - 2 = -222$

$2^{-111} - 2 = -224$

$2^{-112} - 2 = -226$

$2^{-113} - 2 = -228$

$2^{-114} - 2 = -230$

$2^{-115} - 2 = -232$

$2^{-116} - 2 = -234$

$2^{-117} - 2 = -236$

$2^{-118} - 2 = -238$

$2^{-119} - 2 = -240$

$2^{-120} - 2 = -242$

$2^{-121} - 2 = -244$

$2^{-122} - 2 = -246$

$2^{-123} - 2 = -248$

$2^{-124} - 2 = -250$

$2^{-125} - 2 = -252$

$2^{-126} - 2 = -254$

$2^{-127} - 2 = -256$

$2^{-128} - 2 = -258$

$2^{-129} - 2 = -260$

$2^{-130} - 2 = -262$

$2^{-131} - 2 = -264$

$2^{-132} - 2 = -266$

$2^{-133} - 2 = -268$

$2^{-134} - 2 = -270$

$2^{-135} - 2 = -272$

$2^{-136} - 2 = -274$

$2^{-137} - 2 = -276$

$2^{-138} - 2 = -278$

$2^{-139} - 2 = -280$

$2^{-140} - 2 = -282$

$2^{-141} - 2 = -284$

$2^{-142} - 2 = -286$

$2^{-143} - 2 = -288$

$2^{-144} - 2 = -290$

$2^{-145} - 2 = -292$

$2^{-146} - 2 = -294$

$2^{-147} - 2 = -296$

$2^{-148} - 2 = -298$

$2^{-149} - 2 = -300$

$2^{-150} - 2 = -302$

$2^{-151} - 2 = -304$

$2^{-152} - 2 = -306$

$2^{-153} - 2 = -308$

$2^{-154} - 2 = -310$

$2^{-155} - 2 = -312$

$2^{-156} - 2 = -314$

$2^{-157} - 2 = -316$

$2^{-158} - 2 = -318$

$2^{-159} - 2 = -320$

$2^{-160} - 2 = -322$

$2^{-161} - 2 = -324$

$2^{-162} - 2 = -326$

$2^{-163} - 2 = -328$

$2^{-164} - 2 = -330$

$2^{-165} - 2 = -332$

$2^{-166} - 2 = -334$

$2^{-167} - 2 = -336$

$2^{-168} - 2 = -338$

$2^{-169} - 2 = -340$

$2^{-170} - 2 = -342$

$2^{-171} - 2 = -344$

$2^{-172} - 2 = -346$

$2^{-173} - 2 = -348$

$2^{-174} - 2 = -350$

$2^{-175} - 2 = -352$

$2^{-176} - 2 = -354$

$2^{-177} - 2 = -356$

$2^{-178} - 2 = -358$

$2^{-179} - 2 = -360$

$2^{-180} - 2 = -362$

$2^{-181} - 2 = -364$

$2^{-182} - 2 = -366$

$2^{-183} - 2 = -368$

$2^{-184} - 2 = -370$

$2^{-185} - 2 = -372$

2^{-1

```
For more information, please visit wiki.freecode.com.cn.
Press '?' to get help.
Executing the startup file

PC1> show ip
NAME      : PC1[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10050
RHOST:PORT : 127.0.0.1:10051
MTU        : 1500

PC1> ip 192.24.0.2 255.255.255.192 192.24.0.1
Checking for duplicate address...
savePC1 : 192.24.0.2 255.255.255.192 gateway 192.24.0.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip
NAME      : PC1[1]
IP/MASK    : 192.24.0.2/26
GATEWAY    : 192.24.0.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10050
RHOST:PORT : 127.0.0.1:10051
MTU        : 1500

PC1> █
```

```
For more information, please visit wiki.freecode.com.cn.
Press '?' to get help.
Executing the startup file

PC2> show ip
NAME      : PC2[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10052
RHOST:PORT : 127.0.0.1:10053
MTU        : 1500

PC2> ip 192.24.0.66 255.255.255.224 192.24.0.65
Checking for duplicate address...
savePC1 : 192.24.0.66 255.255.255.224 gateway 192.24.0.65

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ip
NAME      : PC2[1]
IP/MASK    : 192.24.0.66/27
GATEWAY    : 192.24.0.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10052
RHOST:PORT : 127.0.0.1:10053
MTU        : 1500

PC2> █
```

```
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 10054
RHOST:PORT : 127.0.0.1:10055
MTU        : 1500

PC3> ip 192.24.0.98 255.255.255.224 192.24.0.97
Checking for duplicate address...
savePC1 : 192.24.0.98 255.255.255.224 gateway 192.24.0.97

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 192.24.0.98/27
GATEWAY    : 192.24.0.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 10054
RHOST:PORT : 127.0.0.1:10055
MTU        : 1500

PC3> [2~
```

```
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10056
RHOST:PORT : 127.0.0.1:10057
MTU        : 1500

PC4> ip 192.24.0.130 255.255.255.240 192.24.0.129
Checking for duplicate address...
savePC1 : 192.24.0.130 255.255.255.240 gateway 192.24.0.129

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 192.24.0.130/28
GATEWAY    : 192.24.0.129
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10056
RHOST:PORT : 127.0.0.1:10057
MTU        : 1500

PC4> 
```

```
PC1 PC2 PC3 PC4 PC5
For more information, please visit wiki.freecode.com.cn.
Press '?' to get help.
Executing the startup file

PC5> show ip

NAME       : PC5[1]
IP/MASK     : 0.0.0.0/0
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:04
LPORT      : 10058
RHOST:PORT  : 127.0.0.1:10059
MTU         : 1500

PC5> ip 192.24.0.146 255.255.255.248 192.24.0.145
Checking for duplicate address...
savePC1 : 192.24.0.146 255.255.255.248 gateway 192.24.0.145

PC5> save
Saving startup configuration to startup.vpc
. done

PC5> show ip

NAME       : PC5[1]
IP/MASK     : 192.24.0.146/29
GATEWAY     : 192.24.0.145
DNS         :
MAC         : 00:50:79:66:68:04
LPORT      : 10058
RHOST:PORT  : 127.0.0.1:10059
MTU         : 1500

PC5> 
```

Configuracion de interfaces en los routers:

```
PC1 PC2 PC3 PC4 PC5 R5 R2 R3 R1 R4
*Mar 1 00:03:20.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R5-00097524(config-if)#exit
R5-00097524(config)#interface fastethernet0/1
R5-00097524(config-if)#ip address 192.24.0.163 255.255.255.252
Bad mask /30 for address 192.24.0.163
R5-00097524(config-if)#ip address 192.24.0.163 255.255.255.252
Bad mask /30 for address 192.24.0.163
R5-00097524(config-if)#ip address 192.24.0.162 255.255.255.252
R5-00097524(config-if)#no shutdown
R5-00097524(config-if)#^
% Invalid input detected at '^' marker.

R5-00097524(config-if)#no shutdown
R5-00097524(config-if)#exit
R5-00097524(config)#interface faste
*Mar 1 00:05:22.923: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:05:23.923: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R5-00097524(config)#interface fastethernet1/0
R5-00097524(config-if)#ip address 192.24.0.170 255.255.255.252
R5-00097524(config-if)#no shutdown
R5-00097524(config-if)#exit
R5-00097524(config)#end
R5-00097524#
*Mar 1 00:05:50.839: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
R5-00097524#
*Mar 1 00:05:51.503: %SYS-5-CONFIG_I: Configured from console by console
*Mar 1 00:05:51.839: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R5-00097524#write memory
Building configuration...
[OK]
R5-00097524#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.1      YES manual up          up
FastEthernet0/1 192.24.0.162    YES manual up          up
FastEthernet1/0 192.24.0.170    YES manual up          up
R5-00097524#
*Mar 1 00:08:20.843: %IP-4-DUPADDR: Duplicate address 192.24.0.162 on FastEthernet0/1, sourced by c202.9ddc.0010
R5-00097524# 
```

```
PC1 PC2 PC3 PC4 PC5 R2 x R3 R1 R4 + - □ x
*Mar 1 00:06:07.027: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:08.027: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2-00097524(config)#interface fastethernet0/1
R2-00097524(config-if)#ip address 192.24.0.154 255.255.255.252
R2-00097524(config-if)#no shutdown
R2-00097524(config-if)#ip
*Mar 1 00:06:34.487: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:06:35.487: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2-00097524(config-if)#exit
R2-00097524(config)#interface fastethernet1/0
^
% Invalid input detected at '^' marker.

R2-00097524(config)#interface fastethernet1/0
R2-00097524(config-if)#ip address 192.24.0.162 255.255.255.252
R2-00097524(config-if)#no shutdown
R2-00097524(config-if)#e
*Mar 1 00:07:11.127: %IP-4-DUPADDR: Duplicate address 192.24.0.162 on FastEthernet1/0, sourced by c205.0898.0001
R2-00097524(config-if)#exit
R2-00097524(config)#
*Mar 1 00:07:13.099: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:07:14.099: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2-00097524(config)#interface fastethernet1/0
R2-00097524(config-if)#ip address 192.24.0.161 255.255.255.252
R2-00097524(config-if)#no shutdown
R2-00097524(config-if)#exit
R2-00097524(config)#end
R2-00097524#write memor
*Mar 1 00:07:51.947: %SYS-5-CONFIG_I: Configured from console by console
R2-00097524#write memory
Building configuration...
[OK]
R2-00097524#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.65     YES manual up          up
FastEthernet0/1 192.24.0.154    YES manual up          up
FastEthernet1/0 192.24.0.161    YES manual up          up
R2-00097524#
```

```
PC1 PC2 PC3 PC4 PC5 R2 R3 x R1 R4 + - □ x
R3(config)#hostname R3-00097524
R3-00097524(config)#interface fastethernet0/0
R3-00097524(config-if)#ip address 192.24.0.97 255.255.255.224
R3-00097524(config-if)#no shutdown
R3-00097524(config-if)#exit
R3-00097524(config)#interface
^
*Mar 1 00:06:33.023: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:34.023: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3-00097524(config)#interface fastethernet 192.24.0.158 255.255.255.252
^
% Invalid input detected at '^' marker.

R3-00097524(config)#interface fastethernet0/1
R3-00097524(config-if)#ip address 192.24.0.158 255.255.255.252
R3-00097524(config-if)#no shutdown
R3-00097524(config-if)#exit
R3-00097524(config)#interface faste
*Mar 1 00:07:16.371: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:07:17.371: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3-00097524(config)#interface fastethernet1/0
R3-00097524(config-if)#ip address 192.24.0.165 255.255.255.252
R3-00097524(config-if)#no shutdown
R3-00097524(config-if)#exit
R3-00097524(config)#
*Mar 1 00:07:50.127: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:07:51.127: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R3-00097524(config)#end
R3-00097524#
*Mar 1 00:07:57.491: %SYS-5-CONFIG_I: Configured from console by console
R3-00097524#write memory
Building configuration...
[OK]
R3-00097524#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.97     YES manual up          up
FastEthernet0/1 192.24.0.158    YES manual up          up
FastEthernet1/0 192.24.0.165    YES manual up          up
R3-00097524#
```

```

R5 R2 R1 R3 R4
% Crashinfo may not be recovered at flash:crashinfo
% This file system device reports an error

Press RETURN to get started!

*Mar 1 00:00:01.467: %SW_VLAN-4-IFS_FAILURE: VLAN manager encountered file oper
ation error: call = ifs_open/read / code = 3588 (No device available)
/ bytes transferred = 0
*Mar 1 00:00:01.599: %SYS-5-CONFIG_I: Configured from memory by console
*Mar 1 00:00:01.675: %LINEPROTO-5-UPDOWN: Line protocol on Interface VoIP-Null0
, changed state to up
*Mar 1 00:00:01.735: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 3700 Software (C3725-ADVENTERPRISEK9-M), Version 12.4(15)T14
, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2010 by Cisco Systems, Inc.
Compiled Tue 17-Aug-10 12:08 by prod_rel_team
*Mar 1 00:00:01.751: %SNMP-5-COLDSTART: SNMP agent on host R3-00097524 is under
going a cold start
*Mar 1 00:00:01.767: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Mar 1 00:00:01.767: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Mar 1 00:00:01.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface IPv6-mpls,
changed state to up
*Mar 1 00:00:02.727: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:00:02.743: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 1 00:00:02.759: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar 1 00:00:03.727: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
*Mar 1 00:00:03.743: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
*Mar 1 00:00:03.759: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
R3-00097524#
```

```

R5 R2 R1 R3 R4
FastEthernet1/0 unassigned YES unset administratively down down
R4#enable
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#hostname R4-00097524
R4-00097524(config)#interface fastethernet0/0
R4-00097524(config-if)#ip address 192.24.0.145 255.255.255.248
R4-00097524(config-if)#no shutdown
R4-00097524(config-if)#exit
R4-00097524(config)#interface faste
*Mar 1 00:02:45.959: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:02:46.959: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4-00097524(config)#interface fastethernet0/1
R4-00097524(config-if)#ip address 192.24.0.166 255.255.255.252
R4-00097524(config-if)#no shutdown
R4-00097524(config-if)#exit
R4-00097524(config)#int
*Mar 1 00:03:11.271: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:12.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4-00097524(config)#interface fastethernet1/0
R4-00097524(config-if)#ip address 192.24.0.169 255.255.255.252
R4-00097524(config-if)#no shutdown
R4-00097524(config-if)#exit
R4-00097524(config)#end
*Mar 1 00:03:37.559: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:03:38.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R4-00097524(config)#end
R4-00097524#write memor
*Mar 1 00:03:38.903: %SYS-5-CONFIG_I: Configured from console by console
R4-00097524#write memory
Building configuration...
[OK]
R4-00097524#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.24.0.145 YES manual up up
FastEthernet0/1 192.24.0.166 YES manual up up
FastEthernet1/0 192.24.0.169 YES manual up up
R4-00097524#
```

Configuracion Dinamica:

```

R5
R2
R1
R3
R4

Enter configuration commands, one per line. End with CNTL/Z.
R5-00097524(config)#router ospf 1
R5-00097524(config-router)#network 192.24.0.64 0.0.0.31 area 0
R5-00097524(config-router)#network 192.24.0.96 0.0.0.31 area 0
R5-00097524(config-router)#network 192.24.0.128 0.0.0.15 area 0
R5-00097524(config-router)#no network 192.24.0.128 0.0.0.15 area 0
R5-00097524(config-router)#no network 192.24.0.128 0.0.0.15 area 0
R5-00097524(config-router)#no network 192.24.0.64 0.0.0.31 area 0
R5-00097524(config-router)# network 192.24.0.0 0.0.0.63 area 0
R5-00097524(config-router)# network 192.24.0.160 0.0.0.3 area 0
R5-00097524(config-router)# network 192.24.0.168 0.0.0.3 area 0
R5-00097524(config-router)#exit
R5-00097524(config)#end
R5-00097524#write memo
*Mar  1 00:05:18.219: %SYS-5-CONFIG_I: Configured from console by console
R5-00097524#write memory
Building configuration...
[OK]
R5-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.24.0.168/30 is directly connected, FastEthernet1/0
C       192.24.0.160/30 is directly connected, FastEthernet0/1
C       192.24.0.0/26 is directly connected, FastEthernet0/0
R5-00097524#
*Mar  1 00:06:23.675: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.161 on FastEthernet0/1 from LOADING to FULL, Loading Done
R5-00097524#
*Mar  1 00:09:48.327: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.169 on FastEthernet1/0 from LOADING to FULL, Loading Done
R5-00097524#
```

```

R2
FastEthernet0/1      192.24.0.154  YES NVRAM  up
FastEthernet1/0     192.24.0.161  YES NVRAM  up
R2-00097524#enable
R2-00097524#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2-00097524(config)#router ospf 1
R2-00097524(config-router)#network 192.24.0.64 0.0.0.31 area 0
R2-00097524(config-router)#network 192.24.0.152 0.0.0.3 area 0
R2-00097524(config-router)#network 192.24.0.160 0.0.0.3 area 0
R2-00097524(config-router)#exit
*Mar 1 00:07:54.095: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.170 on FastEthernet1/0 from LOADING to FULL, Loading Done
R2-00097524(config-router)#exit
R2-00097524(config)#end
R2-00097524#writ
*Mar 1 00:07:57.039: %SYS-5-CONFIG_I: Configured from console by console
R2-00097524#write memory
Building configuration...
[OK]
R2-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 5 subnets, 3 masks
C       192.24.0.152/30 is directly connected, FastEthernet0/1
O       192.24.0.168/30 [110/2] via 192.24.0.162, 00:00:05, FastEthernet1/0
C       192.24.0.160/30 is directly connected, FastEthernet1/0
C       192.24.0.64/27 is directly connected, FastEthernet0/0
O       192.24.0.0/26 [110/11] via 192.24.0.162, 00:00:05, FastEthernet1/0
R2-00097524#
*Mar 1 00:08:58.715: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.157 on FastEthernet0/1 from LOADING to FULL, Loading Done
R2-00097524#
```

```

R1
R1-00097524#enable
R1-00097524#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1-00097524(config)#router ospf 1
R1-00097524(config-router)#network 192.24.0.128 0.0.0.15 area 0
R1-00097524(config-router)#network 192.24.0.152 0.0.0.3 area 0
R1-00097524(config-router)#network 192.24.0.156 0.0.0.3 area 0
R1-00097524(config-router)#
*Mar 1 00:10:38.667: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.161 on FastEthernet0/1 from LOADING to FULL, Loading Done
R1-00097524(config-router)#exit
R1-00097524(config)#end
R1-00097524#write
*Mar 1 00:10:50.659: %SYS-5-CONFIG_I: Configured from console by console
R1-00097524#write memory
Building configuration...
[OK]
R1-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 7 subnets, 4 masks
C       192.24.0.152/30 is directly connected, FastEthernet0/1
C       192.24.0.156/30 is directly connected, FastEthernet1/0
C       192.24.0.128/28 is directly connected, FastEthernet0/0
O       192.24.0.168/30 [110/12] via 192.24.0.154, 00:00:13, FastEthernet0/1
O       192.24.0.160/30 [110/11] via 192.24.0.154, 00:00:13, FastEthernet0/1
O       192.24.0.64/27 [110/20] via 192.24.0.154, 00:00:13, FastEthernet0/1
O       192.24.0.0/26 [110/21] via 192.24.0.154, 00:00:15, FastEthernet0/1
R1-00097524#
*Mar 1 00:11:54.899: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.165 on FastEthernet1/0 from LOADING to FULL, Loading Done
R1-00097524#
```

```

R3-00097524(config)#router ospf 1
R3-00097524(config-router)#network 192.24.0.96 0.0.0.31 area 0
R3-00097524(config-router)#network 192.24.0.156 0.0.0.3 area 0
R3-00097524(config-router)#network 192.24.0.164 0.0.0.3 area 0
R3-00097524(config-router)#exit
*Mar 1 00:10:04.851: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.157 on FastEthernet0/1 from LOADING to FULL, Loading Done
R3-00097524(config-router)#exit
R3-00097524(config)#end
R3-00097524#write memor
*Mar 1 00:10:07.731: %SYS-5-CONFIG_I: Configured from console by console
R3-00097524#write memory
Building configuration...
[OK]
R3-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.0.0/24 is variably subnetted, 9 subnets, 4 masks
O 192.24.0.152/30 [110/20] via 192.24.0.157, 00:00:06, FastEthernet0/1
C 192.24.0.156/30 is directly connected, FastEthernet0/1
O 192.24.0.128/28 [110/20] via 192.24.0.157, 00:00:06, FastEthernet0/1
O 192.24.0.168/30 [110/22] via 192.24.0.157, 00:00:06, FastEthernet0/1
O 192.24.0.160/30 [110/21] via 192.24.0.157, 00:00:06, FastEthernet0/1
C 192.24.0.164/30 is directly connected, FastEthernet1/0
O 192.24.0.64/27 [110/30] via 192.24.0.157, 00:00:08, FastEthernet0/1
C 192.24.0.96/27 is directly connected, FastEthernet0/0
O 192.24.0.0/26 [110/31] via 192.24.0.157, 00:00:08, FastEthernet0/1
R3-00097524#
*Mar 1 00:10:52.015: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.169 on FastEthernet1/0 from LOADING to FULL, Loading Done
R3-00097524#
```

```

R4-00097524(config)#router ospf 1
R4-00097524(config-router)#network 192.24.0.144 0.0.0.7 area 0
R4-00097524(config-router)#network 192.24.0.164 0.0.0.3 area 0
R4-00097524(config-router)#network 192.24.0.164 0.0.0.3 area 0
*Mar 1 00:10:41.803: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.165 on FastEthernet0/1 from LOADING to FULL, Loading Done
R4-00097524(config-router)#network 192.24.0.168 0.0.0.3 area 0
R4-00097524(config-router)#exit
R4-00097524(config)#end
R4-00097524#writ
*Mar 1 00:10:56.871: %SYS-5-CONFIG_I: Configured from console by console
R4-00097524#write memory
Building configuration...
[OK]
R4-00097524#
*Mar 1 00:10:58.427: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.170 on FastEthernet1/0 from LOADING to FULL, Loading Done
R4-00097524#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.0.0/24 is variably subnetted, 10 subnets, 5 masks
O 192.24.0.152/30 [110/21] via 192.24.0.170, 00:00:01, FastEthernet1/0
O 192.24.0.156/30 [110/20] via 192.24.0.165, 00:00:21, FastEthernet0/1
C 192.24.0.144/29 is directly connected, FastEthernet0/0
O 192.24.0.128/28 [110/30] via 192.24.0.165, 00:00:21, FastEthernet0/1
C 192.24.0.168/30 is directly connected, FastEthernet1/0
O 192.24.0.160/30 [110/11] via 192.24.0.170, 00:00:01, FastEthernet1/0
C 192.24.0.164/30 is directly connected, FastEthernet0/1
O 192.24.0.64/27 [110/21] via 192.24.0.170, 00:00:03, FastEthernet1/0
O 192.24.0.96/27 [110/20] via 192.24.0.165, 00:00:23, FastEthernet0/1
O 192.24.0.0/26 [110/11] via 192.24.0.170, 00:00:03, FastEthernet1/0
R4-00097524#
```

Conexión exitosa:

```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
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For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
PC1 : 192.24.0.66 255.255.255.224 gateway 192.24.0.65

PC2> trace 192.24.0.146
Trace to 192.24.0.146, 8 hops max, press Ctrl+C to stop
 1  192.24.0.65    14.554 ms  15.156 ms  15.547 ms
 2  192.24.0.162   45.705 ms  45.746 ms  45.839 ms
 3  192.24.0.169   77.603 ms  76.412 ms  76.056 ms
 4  *192.24.0.146  77.164 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> 
```